

TITLE: GXP and Non-GXP Study Report Review and Approval Form

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Executive summary of study results:

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The ability of several anti-lysophosphatidic acid (anti-LPA) antibodies (both murine, B3, and humanized: LT3015; LT3017; LT3114) to bind to a range of lipids that are structurally and chemically similar to LP was examined in a competition ELISA format. Lipids at varying concentrations were added to the test antibodies in a 3:1 ratio (Lipid:Ab) and then added to plates (100 µl/well in triplicate) coated with 18:0 LPA and the ability of the antibodies to bind to the plate in the presence of potentially competing ligands was assessed.

There was no significant cross-reactivity of any of the antibodies with the lipids tested up to a concentration of 5 µM lipid with the exception of LPA itself.

LPATH APPROVAL

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Comparison of the binding of anti-LPA antibodies to other lipid molecules (cross-reactivity)

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1 SUMMARY

The ability of several anti-lysophosphatidic acid (anti-LPA) antibodies (both murine, B3, and humanized: LT3015; LT3017; LT3114) to bind to a range of lipids that are structurally and chemically similar to LPA was examined in a competition ELISA format. Lipids at varying concentrations were added to the test antibodies in a 3:1 ratio (Lipid:Ab) and then added to plates (100 μ l/well in triplicate) coated with 18:0 LPA and the ability of the antibodies to bind to the plate in the presence of potentially competing ligands was assessed.

There was no significant cross-reactivity of any of the antibodies with the lipids tested up to a concentration of 5 μ M lipid with the exception of LPA itself.

2 INTRODUCTION

The bioactive lipid lysophosphatidic acid (LPA) is an important extracellular signaling molecule in neuropathic pain. LPA and its receptors are responsible for long-lasting mechanical allodynia and thermal hyperalgesia as well as demyelination and upregulation of pain-related protein expression patterns in a variety of animal models. Evidence supports the hypothesis that antagonism of the LPA signaling pathways would benefit the treatment of neuropathic pain. We have developed the first high affinity monoclonal antibodies targeting LPA (anti-LPA mAb) and the studies described here were performed to examine the potential for these antibodies to bind other lipids which are structurally similar to LPA.

3 MATERIALS AND METHODS

3.1 Test Articles

The range of lipids tested in this study is shown in Table 1. Antibodies that were used to examine competitive binding are detailed in Table 2.

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Table 1 Lipids that show similarity to LPA that were used to test for cross-reactivity

Lipid Inventory Id.	“common” name	Chemical name	Vendor	Cat. no.
72	Lysophosphatidic Acid (18:2)	1-Linoleoyl LPA	Echelon	L-0182
76	Lysophosphatidic Acid (C18:1) alkyl	1-(9Z-octadecenyl)-2-hydroxy-sn-glycerol-3-phosphate	Avanti	857230
1	Sphingosine-1-Phosphate	D-erythro-Sphingosine-1-Phosphate	Avanti	860492
3	D-erythro Sphingosine	D-erythro-Sphingosine (Synthetic)	Avanti	860490
5	Sphingosine Phosphocholine	D-erythro-Sphingosine Phosphocholine (Lyso Sphingomyelin)	Avanti	860600
43	Lysophosphatidylcholine (C18:0)	1-Stearoyl-2-Hydroxy-sn-Glycerol-3-Phosphocholine	Avanti	855775
45	Lysophosphatidylethanolamine (C18:0)	1-Stearoyl-2-Hydroxy-sn-Glycerol-3-Phosphoethanolamine	Avanti	856715
47	Lysophosphatidylglycerol (LPG C18:0)	1-Stearoyl-2-Hydroxy-sn-Glycerol-3-[Phospho-rac-1-glycerol]]	Avanti	858124
87	LPS (18:1)	1-Oleoyl-2-hydroxy-sn-glycerol-3-phospho-L-serine	Avanti	858143
90	18:1 Lyso PI	1-Olyoyl-2-Hydroxy-sn-Glycerol-3-Phosphoinositol	Avanti	850100
8	Phosphatidylserine (PS 16:0)	1,2-Dipalmitoyl-sn-Glycerol-3-[Phospho-L-Serine] (Sodium Salt) (DPPS)	Avanti	840037
34	16:0-18:1 PC	1-Palmitoyl-2-Oleoyl-sn-Glycerol-3-Phosphocholine (16:0-18:1)	Avanti	850457
35	Phosphatidylglycerol (18:2 PG)	1,2-Dilinoleoyl-sn-Glycerol-3-[Phospho-rac-(1-glycerol)]	Avanti	840485
36	Phosphatidic Acid 16:0-18:1 PA	1-Palmitoyl-2-Oleoyl-sn-Glycerol-3-Phosphate (16:0-18:1)	Avanti	840857
88	18:1 (Δ 9-trans) PE	1,2-dieladoyl-sn-glycerol-3-phosphoethanolamine	Avanti	850726
89	11:0 PC	1,2-diundecanoyl-sn-glycerol-3-phosphocholine	Avanti	850330
14	PAF	1-Alkyl-2-Acetoyleyl-sn-Glycerol-3-Phosphocholine	Avanti	840009
16	cyclic LPA (18:1)	1-Oleoyl-sn-Glycerol-2-3-Cyclic-Phosphate	Avanti	857328
19	Lyso PAF	1-Alkyl-2-Hydroxy-sn-Glycerol-3-	Avanti	840008

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Table 1 Lipids that show similarity to LPA that were used to test for cross-reactivity

		Phosphocholine		
86	BrP-LPA	1-Bromo-3(S)-hydroxy-4-(palmitoyloxy) butyl] phosphonate	Echelon	L-7416
50	Diacylglycerol (18:1 DG)	1,2-dioleoyl- <i>sn</i> -glycerol	Avanti	800811
85	C12-ether LBPA	(<i>R,R</i>)-2,2'-Bisdodecyl-LBPA ammonium salt	Echelon	L-B120e
91	2-AG	2-arachidonoyl glycerol	Avanti	870450
		Glycerol phosphate (40%)	Fluka	3391

Table 2 Humanized and murine antibodies used in these experiments

Antibody	Part Number	Target	Lot Number	Concentration (mg/mL)	Dilution (µg/mL)
Humanized anti-LPA	LT 3015	LPA	LR81055	19.3 mg/ml	0.5 µg/ml
Humanized anti-LPA	LT 3017	LPA	LP91090-42	0.91 mg/ml	0.5 µg/ml
Humanized anti-LPA	LT 3114	LPA	LP91183	1.54 mg/ml	0.5 µg/ml
Murine anti-LPA	504B3	LPA	LP91172	11.4 mg/ml	0.5 µg/ml

3.2 ELISA methodology

The specificity of the humanized antibody was determined by competition ELISA. 18:0 LPA coating material is diluted to 0.33 µg/ml in carbonate buffer (100 mM NaHCO₃, 33.6 mM Na₂CO₃, pH 9.5). Plates are coated with 100 µl/well of this coating solution and incubated at 37°C for 1 hour. The plates are then washed 4X with PBS (100 mM Na₂HPO₄, 20 mM KH₂PO₄, 27 mM KCl, 1.37 mM NaCl, pH 7.4) and blocked with 150 µl/well PBS + 1% BSA + 0.1% tween-20 for 1 hour at room temperature. The humanized anti-LPA antibody variants and the murine B3 antibody were tested against lipids that are structurally and chemically similar to LPA at 5 µM, 2.5 µM, 1.25 µM, 0.625 µM, and 0.0 µM. The antibodies are diluted to 0.5 µg/ml in PBS +0.1 % tween-20 and combined with the lipid samples at a 1:3 ratio of antibody to sample on a non-binding plate (recommended 100 µl Ab

and 300 µl samples/standards, enough to run in triplicate). The plates are washed 4X with PBS and then incubated for 1 hour at room temperature with 100 µl/well of the primary antibody combined with the samples. Next the plates are washed 4X with PBS and then incubated for 1 hour at room temperature with 100 µl/well of HRP-conjugated goat anti-human or murine antibody diluted 1:20,000 in PBS + 1% BSA + 0.1% tween-20. Again the plates are washed 4X with PBS and developed using 100 µl/well TMB substrate at 4°C. After 8 minutes, the reaction is stopped with 1M H₂SO₄, 100 µl/well. The optical density (OD) was measured at 450 nm using a Thermo Multiskan EX. Raw data were transferred to GraphPad software for analysis of competition curves. Values for percent inhibition at 2.5 µM lipid were also calculated for each lipid using Microsoft Excel.

4 RESULTS

Competition curves for all lipids tested are shown. Individual data tables for these lipids are detailed in Appendix 1. Data showing the percent inhibition of all lipids at 2.5µM is summarized in Table 3. Overall, as expected the only lipid that demonstrated significant competition with the coating material (18:0 LPA) for binding was LPA 18:2. 18:1 alkyl-LPA also showed a low level of inhibition in these assay but none of the remaining lipids showed a significant level of cross-reactivity as measured by competition for plated 18:0 LPA.

Table 3 Percent Inhibition of antibody binding to 18:0 LPA at 2.5 μM lipid				
Lipid	LT 3015	LT3017	LT3114	B3
LPA 18:2	44%	47%	61%	71%
LPA C18:1	12%	12%	20%	30%
S1P	-1%	0%	-10%	4%
sphingosine	5%	8%	7%	3%
sphingosine phosphocholine	3%	3%	-1%	2%
Lysophosphatidylcholine (C18:0)	2%	2%	2%	-2%
Lysophosphatidylethanolamine (C18:0)	5%	1%	-3%	-2%
Lysophosphatidylglycerol (LPG C18:0)	7%	3%	7%	8%
LPS C18:1	3%	0%	-1%	2%
18:1 Lyso PI	-1%	-3%	0%	3%
phosphatidylserine (PS 16:0)	0%	2%	1%	3%
Phosphocholine(16:0-18:1)	0%	3%	0%	1%
Phosphatidylglycerol (18:2 PG)	8%	14%	6%	0%
Phosphatidic Acid 16:0-18:1 PA	1%	0%	1%	0%
18:1 (Δ 9-trans) PE	6%	6%	0%	6%
11:0 PC	2%	1%	1%	-1%
PAF	12%	-6%	-5%	4%
cyclic LPA (18:1)	-13%	9%	9%	16%
Lyso PAF	2%	7%	0%	6%
BrP-LPA	2%	-1%	-5%	6%
Diacylglycerol (18:1 DG)	-3%	-1%	-5%	-5%
C12-ether LBPA	-5%	1%	-1%	2%
2-AG	7%	3%	8%	7%
glycerol phosphate	5%	3%	1%	3%

5 CONCLUSIONS

There is no significant cross-reactivity with any of the lipids tested up to 5 μ M with the exception of LPA.

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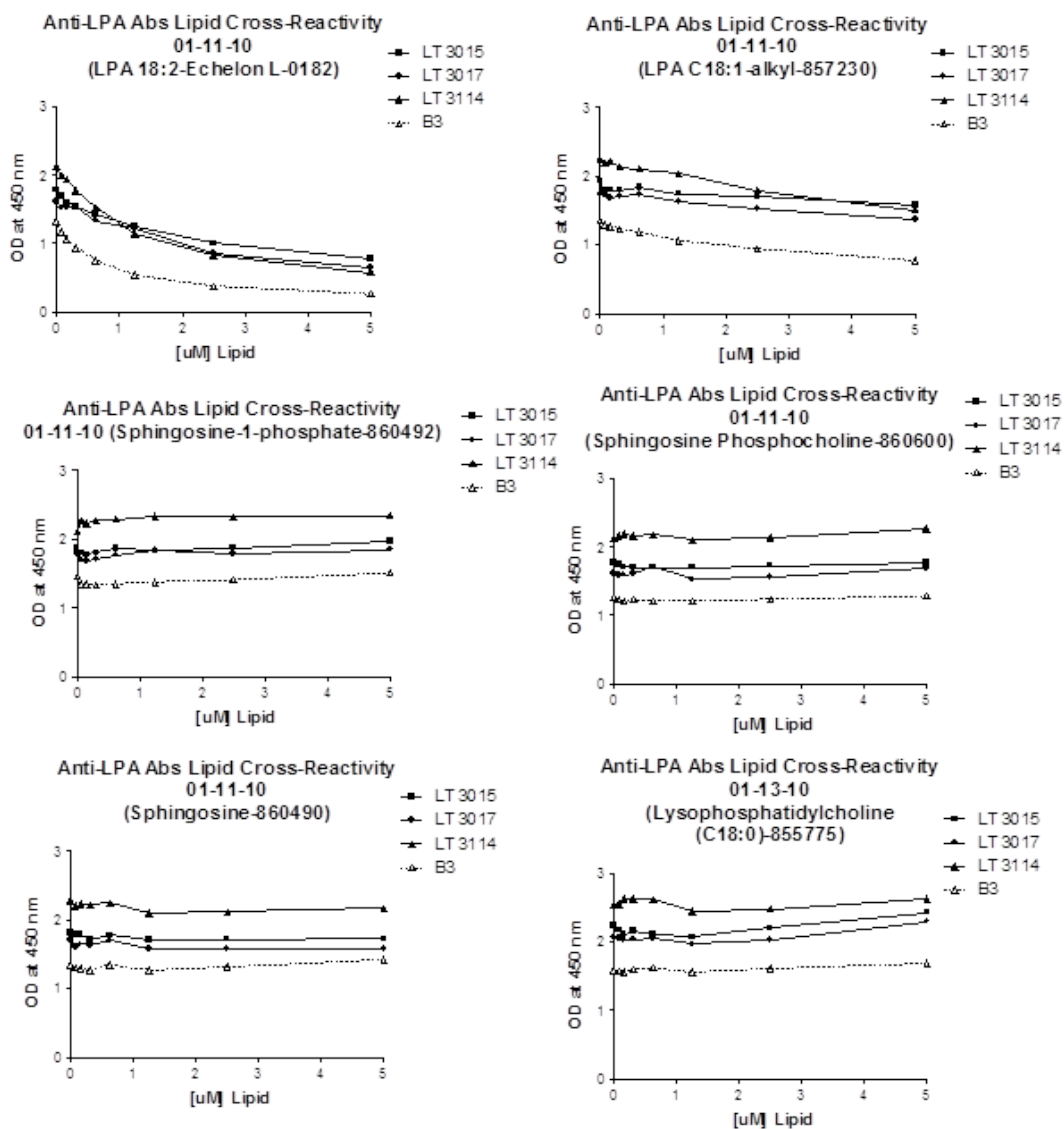
6 DATA ARCHIVE

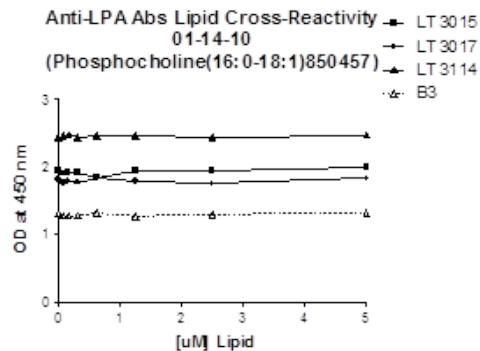
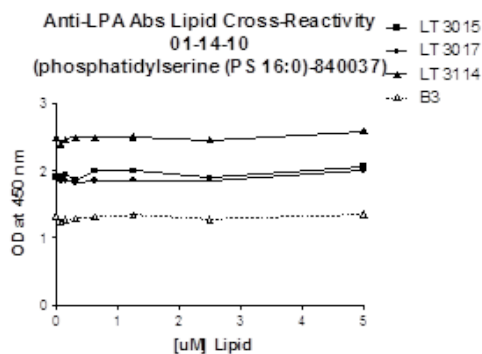
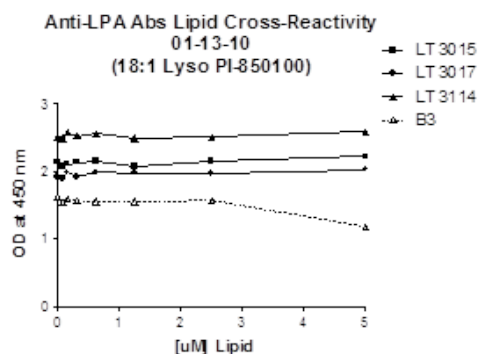
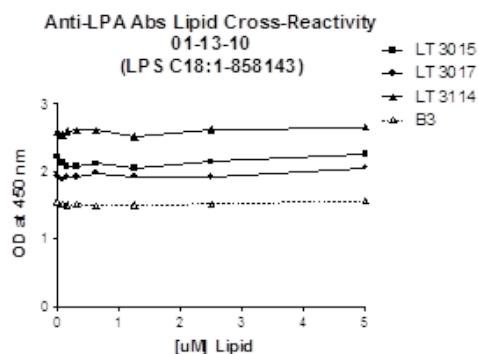
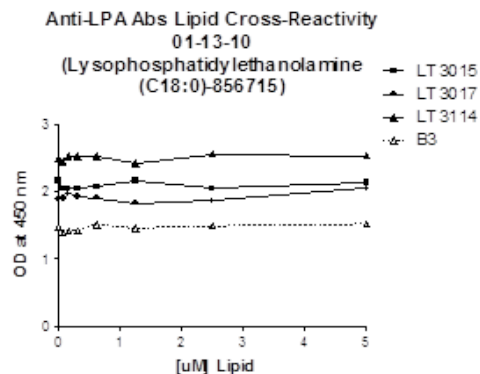
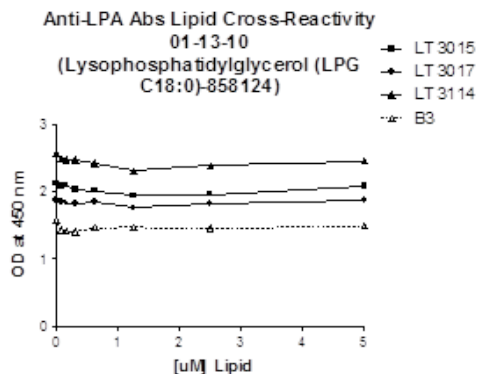
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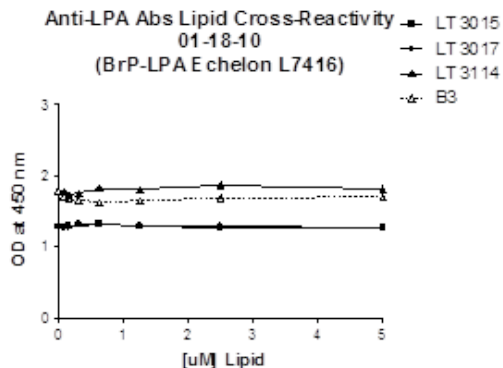
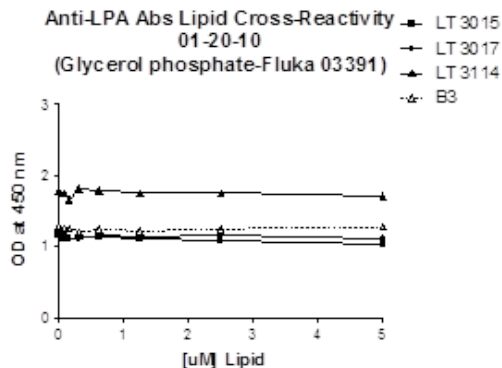
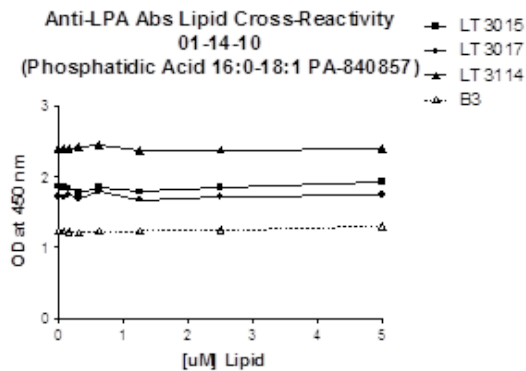
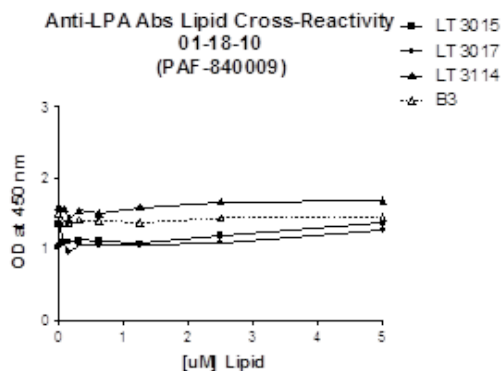
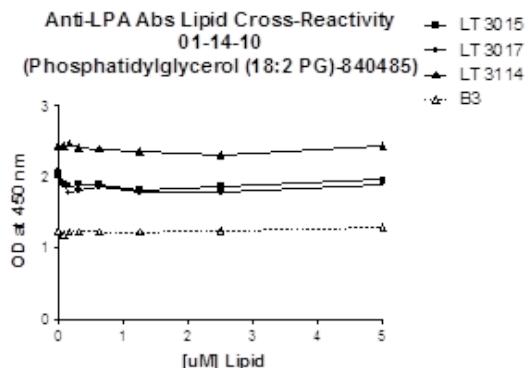
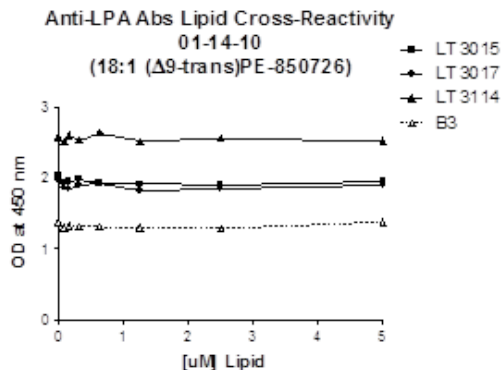
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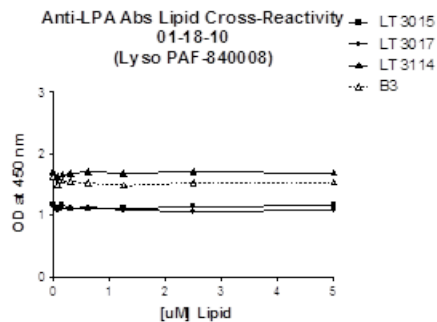
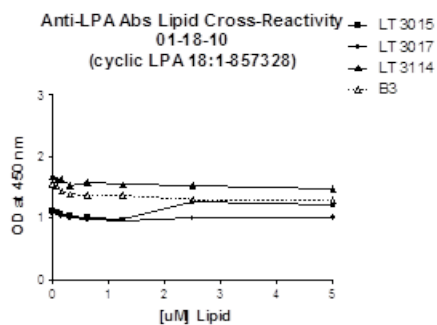
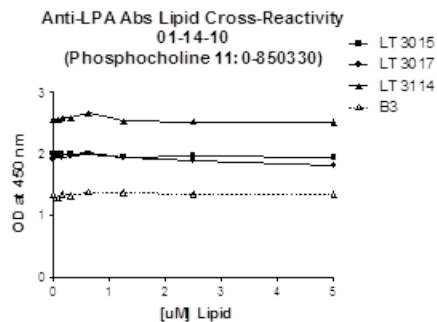
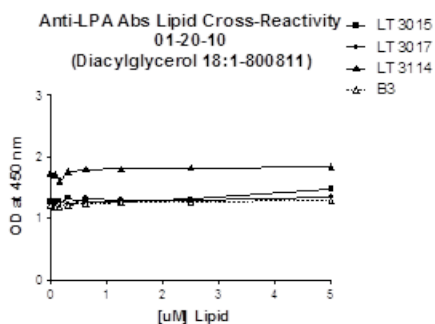
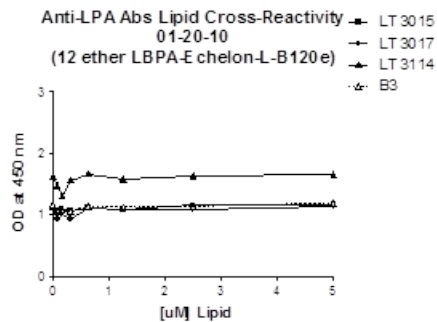
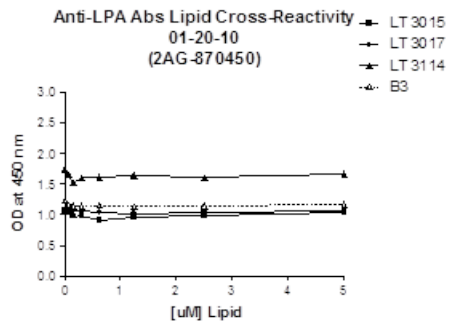
7 APPENDICES

Appendix 1 Competition curves of all lipids tested









Appendix 2 Raw Data Compilation

LPA 18:2	LT 3015			LT 3017			LT 3114			B3		
5	0.785	0.779	0.755	0.688	0.599	0.657	0.607	0.567	0.573	0.273	0.245	0.275
2.5	1.028	1	0.992	0.879	0.839	0.867	0.85	0.821	0.833	0.383	0.365	0.385
1.25	1.248	1.229	1.257	1.413	1.139	1.108	1.109	1.154	1.151	0.53	0.542	0.541
0.625	1.439	1.423	1.432	1.329	1.321	1.363	1.534	1.501	1.533	0.759	0.752	0.76
0.3125	1.577	1.529	1.516	1.628	1.48	1.487	1.768	1.768	1.767	0.94	0.938	0.934
0.15625	1.616	1.611	1.594	1.571	1.49	1.54	1.916	1.924	1.983	1.071	1.058	1.074
0.078125	1.742	1.669	1.656	1.541	1.516	1.512	1.946	1.985	2.036	1.176	1.172	1.155
0	1.799	1.781	1.785	1.578	1.654	1.618	2.118	2.063	2.191	1.331	1.298	1.344

LPA C18:1	LT 3015			LT 3017			LT 3114			B3		
5	1.633	1.557	1.561	1.353	1.408	1.331	1.497	1.57	1.451	0.749	0.766	0.791
2.5	1.719	1.707	1.68	1.568	1.546	1.481	1.766	1.842	1.77	0.943	0.94	0.946
1.25	1.754	1.746	1.731	1.601	1.679	1.606	1.993	1.981	2.119	1.074	1.08	1.026
0.625	1.845	1.828	1.842	1.75	1.751	1.712	2.114	2.099	2.092	1.202	1.171	1.168
0.3125	1.806	1.789	1.795	1.713	1.725	1.702	2.078	2.189	2.138	1.212	1.244	1.224
0.15625	1.816	1.786	1.785	1.67	1.732	1.636	2.214	2.219	2.217	1.262	1.281	1.263
0.078125	1.796	1.796	1.792	1.726	1.715	1.742	2.173	2.173	2.212	1.3	1.29	1.28
0	2.028	1.882	1.883	1.762	1.767	1.691	2.234	2.272	2.237	1.36	1.326	1.366

S1P	LT 3015			LT 3017			LT 3114			B3		
5	1.996	1.97	1.931	1.806	1.871	1.873	2.473	2.172	2.387	1.452	1.522	1.556
2.5	1.874	1.881	1.872	1.804	1.804	1.759	2.377	2.298	2.303	1.403	1.426	1.395
1.25	1.786	1.841	1.851	1.909	1.882	1.759	2.346	2.298	2.356	1.374	1.385	1.367
0.625	1.836	1.871	1.883	1.796	1.731	1.794	2.245	2.305	2.329	1.358	1.36	1.325
0.3125	1.816	1.833	1.772	1.663	1.756	1.719	2.293	2.293	2.235	1.312	1.35	1.324
0.15625	1.732	1.831	1.771	1.7	1.731	1.636	2.238	2.238	2.228	1.35	1.351	1.342
0.078125	1.775	1.839	1.797	1.686	1.758	1.654	2.267	2.271	2.287	1.316	1.374	1.358
0	1.915	1.868	1.802	1.762	1.797	1.797	2.106	2.138	2.103	1.432	1.479	1.475

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Sphingosine	LT 3015			LT 3017			LT 3114			B3		
5	1.748	1.711	1.701	1.55	1.602	1.602	2.142	2.18	2.173	1.39	1.466	1.407
2.5	1.721	1.721	1.704	1.637	1.508	1.571	2.084	2.109	2.137	1.3	1.326	1.308
1.25	1.671	1.702	1.736	1.616	1.549	1.554	2.053	2.115	2.113	1.228	1.259	1.318
0.625	1.774	1.772	1.769	1.719	1.691	1.676	2.276	2.268	2.191	1.338	1.337	1.339
0.3125	1.735	1.69	1.689	1.608	1.663	1.623	2.235	2.205	2.203	1.246	1.237	1.314
0.15625	1.798	1.784	1.774	1.643	1.653	1.633	2.197	2.232	2.235	1.28	1.286	1.286
0.078125	1.874	1.703	1.767	1.631	1.578	1.615	2.151	2.205	2.232	1.242	1.31	1.372
0	1.873	1.776	1.78	1.733	1.711	1.695	2.278	2.278	2.281	1.275	1.383	1.377
SPC	LT 3015			LT 3017			LT 3114			B3		
5	1.82	1.785	1.729	1.692	1.726	1.662	2.264	2.273	2.249	1.257	1.248	1.352
2.5	1.773	1.728	1.688	1.552	1.575	1.555	2.146	2.083	2.189	1.184	1.233	1.291
1.25	1.717	1.705	1.684	1.541	1.55	1.48	2.082	2.08	2.132	1.2	1.215	1.225
0.625	1.7	1.688	1.715	1.943	1.622	1.557	2.18	2.156	2.195	1.156	1.234	1.237
0.3125	1.752	1.701	1.659	1.638	1.644	1.571	2.198	2.163	2.139	1.216	1.174	1.313
0.15625	1.768	1.703	1.663	1.599	1.615	1.58	2.213	2.165	2.176	1.186	1.191	1.254
0.078125	1.759	1.76	1.72	1.644	1.617	1.504	2.157	2.157	2.173	1.195	1.25	1.259
0	1.857	1.744	1.736	1.589	1.63	1.623	2.117	2.134	2.123	1.178	1.287	1.329
LPC 18:0	LT 3015			LT 3017			LT 3114			B3		
5	2.488	2.392	2.4	2.285	2.344	2.267	2.645	2.604	2.645	1.631	1.665	1.756
2.5	2.347	2.148	2.106	1.987	2.098	1.987	2.477	2.477	2.481	1.568	1.576	1.684
1.25	2.098	2.036	2.075	1.927	1.988	1.967	2.411	2.431	2.476	1.547	1.539	1.58
0.625	2.14	2.07	2.114	1.996	2.057	2.085	2.642	2.575	2.642	1.588	1.594	1.684
0.3125	2.263	2.07	2.137	2.018	2.042	2.046	2.609	2.669	2.598	1.565	1.557	1.663
0.15625	2.119	2.125	2.069	2.002	2.072	2.021	2.633	2.604	2.633	1.538	1.55	1.599
0.078125	2.157	2.095	2.26	1.982	2.125	2.04	2.642	2.493	2.493	1.499	1.529	1.691
0	2.397	2.137	2.169	1.973	2.133	2.111	2.542	2.558	2.522	1.539	1.564	1.63

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LPE 18:0	LT 3015			LT 3017			LT 3114			B3		
5	2.186	2.145	2.048	1.966	1.974	2.186	2.538	2.522	2.538	1.477	1.528	1.555
2.5	2.034	2.033	2.064	1.816	1.795	1.981	2.553	2.623	2.487	1.445	1.5	1.524
1.25	1.998	2.19	2.292	1.871	1.757	1.848	2.4	2.439	1.642	1.439	1.464	1.462
0.625	2.103	2.071	2.022	1.923	1.917	1.884	2.468	2.578	2.503	1.552	1.499	1.475
0.3125	2.026	2.076	2.02	1.843	1.917	1.986	2.519	2.538	2.505	1.442	1.425	1.382
0.15625	2.062	2.025	2.047	1.939	1.95	2.01	2.558	2.49	2.508	1.382	1.441	1.412
0.078125	2.059	2.042	2.037	1.879	1.887	1.915	2.427	2.457	2.444	1.407	1.383	1.373
0	2.159	2.125	2.168	1.872	1.893	1.895	2.465	2.524	2.465	1.446	1.466	1.483
LPG 18:0	LT 3015			LT 3017			LT 3114			B3		
5	2.131	2.049	2.072	1.86	1.861	1.895	2.47	2.437	2.451	1.472	1.444	1.548
2.5	2.013	1.929	1.931	1.793	1.829	1.828	2.353	2.376	2.422	1.407	1.441	1.487
1.25	1.929	1.926	1.95	1.763	1.744	1.125	2.291	1.001	2.318	1.389	1.444	1.568
0.625	2.077	1.951	1.997	1.838	1.85	1.837	2.37	2.452	1.811	1.433	1.446	1.497
0.3125	2.099	1.968	1.996	1.756	1.812	1.881	2.479	2.466	2.445	1.352	1.418	1.421
0.15625	2.118	2.05	2.088	1.837	1.811	1.828	2.468	2.443	2.49	1.353	1.429	1.46
0.078125	2.156	2.072	2.015	1.838	1.838	1.843	2.475	2.47	2.489	1.399	1.412	1.47
0	2.193	2.088	2.058	1.893	1.854	1.873	2.514	2.535	2.639	1.488	1.638	1.562
LPS C18:1	LT 3015			LT 3017			LT 3114			B3		
5	2.295	2.288	2.186	2.111	1.989	2.06	2.696	2.641	2.619	1.536	1.529	1.615
2.5	2.143	2.162	2.101	1.951	1.926	1.888	2.608	2.615	2.608	1.474	1.492	1.575
1.25	2.038	1.818	0.906	0.928	1.604	1.927	2.586	2.555	2.393	1.471	1.485	1.532
0.625	2.105	2.083	2.14	1.96	1.991	1.977	2.593	2.629	2.588	1.442	1.514	1.521
0.3125	2.101	2.049	2.061	1.908	1.936	1.911	2.595	2.625	2.589	1.487	1.501	1.547
0.15625	2.139	2.087	1.985	1.904	1.943	1.881	2.596	2.562	2.608	1.45	1.522	1.509
0.078125	2.175	2.109	2.099	1.87	1.867	1.904	2.569	2.486	2.563	1.485	1.527	1.529
0	2.328	2.152	2.146	1.769	2.055	1.953	2.616	2.533	2.578	1.511	1.569	1.574

LPI 18:1	LT 3015			LT 3017			LT 3114			B3		
5	2.205	2.205	2.242	2.085	1.995	2.003	2.603	2.59	2.54	1.156	1.183	1.187
2.5	2.147	2.157	2.129	1.976	1.953	1.953	2.509	2.509	2.483	1.522	1.555	1.628
1.25	2.066	2.09	2.036	2.008	1.951	1.97	2.438	2.513	2.513	1.54	1.524	1.577
0.625	2.122	2.188	2.139	1.967	1.968	1.999	2.508	2.591	2.553	1.529	1.543	1.56
0.3125	2.155	2.11	2.139	1.908	1.947	1.894	2.491	2.57	2.51	1.545	1.56	1.582
0.15625	2.118	2.114	2.082	2.009	1.958	1.968	2.565	2.576	2.582	1.56	1.543	1.653
0.078125	2.074	2.097	2.06	1.934	1.885	1.859	2.46	2.414	2.59	1.521	1.533	1.568
0	2.178	2.159	2.055	1.909	1.917	1.912	2.53	2.489	2.484	1.586	1.606	1.649

PS 16:0	LT 3015			LT 3017			LT 3114			B3		
5	2.027	2.074	2.109	2.109	1.919	1.982	2.586	2.595	2.586	1.353	1.301	1.391
2.5	1.862	1.92	1.894	1.897	1.894	1.793	2.438	2.464	2.464	1.227	1.303	1.306
1.25	1.961	2.029	1.992	1.906	1.867	1.804	2.469	2.476	2.551	1.272	1.33	1.427
0.625	2.013	1.966	2.015	1.805	1.826	1.903	2.45	2.525	2.499	1.252	1.364	1.314
0.3125	1.799	1.841	1.919	1.8	1.821	1.841	2.526	2.452	2.494	1.314	1.224	1.333
0.15625	1.914	1.972	1.936	1.839	1.869	1.839	2.502	2.424	2.465	1.249	1.313	1.266
0.078125	1.92	1.908	1.899	1.812	1.884	1.806	2.361	2.403	2.414	1.228	1.194	1.303
0	1.918	1.911	1.87	1.786	1.913	2.01	2.478	2.485	2.485	1.291	1.277	1.402

PC 16:0-18:1	LT 3015			LT 3017			LT 3114			B3		
5	1.993	1.971	2.007	1.814	1.838	1.823	2.445	2.454	2.508	1.294	1.321	1.348
2.5	1.953	1.92	1.941	1.768	1.738	1.749	2.397	2.448	2.467	1.23	1.314	1.336
1.25	1.958	1.908	1.946	1.769	1.772	1.827	2.49	2.435	2.457	1.226	1.287	1.296
0.625	1.876	1.869	1.796	1.814	1.847	1.814	2.442	2.482	2.466	1.327	1.321	1.289
0.3125	1.915	1.913	1.921	1.76	1.761	1.777	2.365	2.421	2.515	1.278	1.314	1.237
0.15625	1.956	1.904	1.884	1.803	1.788	1.763	2.486	2.433	2.518	1.225	1.307	1.293
0.078125	1.941	1.913	1.858	1.767	1.774	1.728	2.416	2.444	2.507	1.242	1.297	1.316
0	1.975	1.952	1.873	1.819	1.816	1.804	2.439	2.384	2.469	1.266	1.319	1.322

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PG 18:2	LT 3015			LT 3017			LT 3114			B3		
5	2.017	1.921	1.939	1.816	1.919	1.959	2.441	2.432	2.41	1.27	1.328	1.258
2.5	1.902	1.861	1.825	1.757	1.828	1.763	2.301	2.328	2.282	1.216	1.221	1.273
1.25	1.846	1.787	1.788	1.7	1.812	1.802	2.336	2.405	2.332	1.201	1.236	1.217
0.625	1.893	1.854	1.91	1.775	1.91	1.893	2.41	2.407	2.362	1.212	1.227	1.232
0.3125	1.899	1.881	1.907	1.737	1.914	1.758	2.376	2.434	2.408	1.229	1.246	1.203
0.15625	1.912	1.912	1.797	1.754	1.826	1.741	2.444	2.504	2.46	1.202	1.253	1.236
0.078125	1.821	1.941	1.936	1.89	1.864	1.873	2.475	2.396	2.423	1.172	1.188	1.191
0	1.957	2.096	2.021	2.052	2.14	2.043	2.524	2.409	2.382	1.224	1.236	1.249

PA 16:0-18:1	LT 3015			LT 3017			LT 3114			B3		
5	1.907	1.935	1.928	1.721	1.712	1.813	2.366	2.452	2.382	1.226	1.319	1.366
2.5	1.842	1.877	1.831	1.727	1.687	1.735	2.332	2.479	2.335	1.187	1.204	1.341
1.25	1.77	1.816	1.775	1.665	1.647	1.714	2.264	2.458	2.389	1.189	1.171	1.343
0.625	1.825	1.893	1.845	1.762	1.826	1.796	2.403	2.496	2.437	1.233	1.17	1.291
0.3125	1.799	1.735	1.805	1.621	1.721	1.754	2.373	2.497	2.397	1.186	1.152	1.297
0.15625	1.838	1.829	1.839	1.734	1.738	1.753	2.342	2.456	2.408	1.196	1.16	1.317
0.078125	1.856	1.86	1.835	1.678	1.647	1.798	2.298	2.489	2.408	1.193	1.204	1.325
0	1.834	1.867	1.898	1.654	1.715	1.772	2.39	2.39	2.406	1.219	1.189	1.315

PE 18:1	LT 3015			LT 3017			LT 3114			B3		
5	1.958	1.881	2.011	1.93	1.904	1.824	2.538	2.488	2.532	1.408	1.362	1.362
2.5	1.838	1.84	2.019	1.841	1.845	1.82	2.486	2.667	2.512	1.332	1.217	1.296
1.25	1.879	1.878	1.972	1.789	1.808	1.849	2.55	2.437	2.544	1.328	1.267	1.302
0.625	1.903	1.904	1.974	1.957	1.934	1.873	2.541	2.676	2.683	1.316	1.306	1.296
0.3125	1.949	1.969	2.036	1.867	1.951	1.875	2.543	2.495	2.553	1.349	1.286	1.307
0.15625	1.913	1.915	1.995	1.867	1.826	1.842	2.586	2.586	2.634	1.363	1.292	1.307
0.078125	1.904	1.946	1.951	1.869	1.886	1.815	2.487	2.532	2.487	1.339	1.278	1.264
0	2.042	1.951	2.094	1.977	1.951	1.954	2.63	2.534	2.534	1.386	1.339	1.38

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PC 11:0	LT 3015			LT 3017			LT 3114			B3		
5	1.986	1.911	1.919	1.833	1.869	1.741	2.471	2.541	2.513	1.362	1.376	1.269
2.5	1.913	1.972	1.995	1.923	1.878	1.862	2.494	2.581	2.489	1.349	1.381	1.299
1.25	1.922	1.99	1.945	1.901	1.966	1.945	2.503	2.547	2.536	1.359	1.429	1.306
0.625	1.984	2.025	2.021	1.946	2.015	2.049	2.666	2.717	2.592	1.411	1.388	1.339
0.3125	2.021	1.997	1.994	1.896	2.021	1.997	2.571	2.594	2.571	1.347	1.332	1.263
0.15625	1.971	2.039	1.981	1.939	1.91	1.937	2.637	2.539	2.561	1.32	1.437	1.276
0.078125	1.995	1.978	2.04	1.971	1.971	1.927	2.567	2.579	2.484	1.315	1.296	1.224
0	1.999	2.026	2.002	1.899	1.929	1.908	2.583	2.531	2.531	1.346	1.358	1.28
PAF	LT 3015			LT 3017			LT 3114			B3		
5	1.468	1.348	1.306	1.387	1.148	1.248	1.704	1.658	1.687	1.443	1.444	1.451
2.5	1.288	1.176	1.085	1.112	1.05	1.115	1.665	1.652	1.676	1.421	1.452	1.425
1.25	1.061	1.09	1.066	1.023	1.05	1.062	1.612	1.612	1.539	1.376	1.306	1.409
0.625	1.162	1.091	1.096	1.029	1.034	1.1	1.338	1.55	1.606	1.367	1.428	1.405
0.3125	1.154	1.09	1.119	1.109	1.064	1.027	1.563	1.525	1.542	1.35	1.438	1.413
0.15625	1.211	1.074	1.021	0.937	0.932	0.976	1.397	1.408	1.465	1.302	1.417	1.336
0.078125	1.124	1.723	1.074	1.085	1.024	1.086	1.526	1.619	1.549	1.391	1.357	1.324
0	1.418	1.456	1.16	1.033	1.031	1.036	1.612	1.585	1.575	1.493	1.402	1.593
cyclic LPA	LT 3015			LT 3017			LT 3114			B3		
5	1.118	1.386	1.14	1.003	0.995	1.036	1.438	1.482	1.482	1.249	1.271	1.345
2.5	1.044	1.738	0.978	1.005	0.967	0.999	1.544	1.5	1.509	1.266	1.326	1.299
1.25	1.012	0.943	0.955	0.968	0.936	0.991	1.52	1.575	1.538	1.34	1.383	1.369
0.625	1.011	1.009	0.977	0.978	0.974	0.949	1.509	1.626	1.612	1.372	1.331	1.391
0.3125	0.986	1.633	1.07	1.029	0.97	0.999	1.502	1.55	1.533	1.372	1.398	1.389
0.15625	1.024	1.067	1.059	1.028	0.997	1.036	1.666	1.633	1.605	1.399	1.446	1.452
0.078125	1.08	1.131	1.087	1.082	1.058	1.071	1.615	1.624	1.625	1.466	1.524	1.53
0	1.069	1.131	1.14	1.096	1.081	1.084	1.628	1.696	1.687	1.559	1.542	1.55

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Lyso PAF	LT 3015			LT 3017			LT 3114			B3		
5	1.133	1.182	1.161	1.051	1.089	1.1	1.7	1.676	1.699	1.49	1.526	1.579
2.5	1.145	1.153	1.139	1.055	1.049	1.09	1.693	1.716	1.694	1.559	1.46	1.55
1.25	1.071	1.158	1.099	1.076	1.061	1.111	1.709	1.641	1.704	1.474	1.5	1.459
0.625	1.096	1.084	1.109	1.116	1.117	1.134	1.758	1.695	1.674	1.469	1.521	1.57
0.3125	1.065	1.113	1.126	1.068	1.129	1.158	1.706	1.666	1.654	1.518	1.553	1.568
0.15625	1.148	1.161	1.162	1.075	1.103	1.117	1.685	1.632	1.672	1.549	1.549	1.58
0.078125	1.098	1.158	1.119	1.077	1.037	1.095	1.617	1.605	1.648	1.473	1.496	1.521
0	1.21	1.151	1.159	1.126	1.147	1.163	1.721	1.651	1.741	1.583	1.6	1.703

BrP-LPA	LT 3015			LT 3017			LT 3114			B3		
5	1.276	1.262	1.262	1.263	1.305	1.264	1.839	1.773	1.809	1.673	1.662	1.752
2.5	1.231	1.248	1.289	1.231	1.314	1.345	1.947	1.766	1.886	1.644	1.661	1.744
1.25	1.246	1.314	1.285	1.268	1.299	1.298	1.833	1.807	1.763	1.609	1.631	1.696
0.625	1.287	1.322	1.346	1.39	1.336	1.26	1.857	1.811	1.798	1.574	1.627	1.658
0.3125	1.305	1.292	1.296	1.287	1.365	1.31	1.761	1.745	1.744	1.615	1.682	1.661
0.15625	1.293	1.279	1.289	1.297	1.285	1.226	1.742	1.715	1.714	1.663	1.686	1.66
0.078125	1.284	1.26	1.274	1.266	1.28	1.271	1.732	1.793	1.793	1.618	1.699	1.787
0	1.257	1.252	1.32	1.289	1.262	1.283	1.803	1.734	1.786	1.664	1.717	1.977
DAG	LT 3015			LT 3017			LT 3114			B3		
5	1.569	1.469	1.4	1.35	1.354	1.311	1.826	1.85	1.839	1.286	1.25	1.342
2.5	1.434	1.218	1.278	1.271	1.329	1.307	1.799	1.841	1.819	1.257	1.27	1.258
1.25	1.391	1.217	1.195	1.326	1.269	1.288	1.803	1.843	1.752	1.243	1.266	1.254
0.625	1.28	1.263	1.245	1.318	1.32	1.305	1.795	1.82	1.757	1.223	1.241	1.24
0.3125	1.439	1.227	1.318	2.078	1.223	1.225	1.75	1.804	1.727	1.196	1.227	1.215
0.15625	1.351	1.265	1.216	1.404	1.212	1.223	1.586	1.622	1.604	1.205	1.195	1.175
0.078125	1.502	1.212	1.127	1.198	1.255	1.239	1.668	1.716	1.743	1.192	1.183	1.2
0	1.497	1.053	1.266	1.318	1.273	1.261	1.649	1.756	1.782	1.227	1.19	1.201

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LBPA	LT 3015			LT 3017			LT 3114			B3		
5	1.197	1.178	1.134	1.185	1.132	1.093	1.688	1.651	1.628	1.168	1.151	1.245
2.5	1.131	1.166	1.154	1.082	1.12	1.065	1.647	1.63	1.606	1.121	1.148	1.145
1.25	1.076	1.085	1.055	1.065	1.103	1.082	1.539	1.595	1.583	1.122	1.143	1.107
0.625	1.069	1.119	1.167	1.092	1.143	1.114	1.679	1.631	1.678	1.162	1.124	1.102
0.3125	1.081	1.043	1.068	1.021	1.043	0.709	1.498	1.548	1.64	1.094	1.032	1.048
0.15625	1.085	1.079	1.104	1.033	1.071	0.869	1.211	1.338	1.358	1.057	1.103	1.046
0.078125	0.981	1.038	1.023	0.92	1.052	0.813	1.543	1.344	1.538	1.021	1.11	1.089
0	1.11	1.104	1.077	1.101	1.113	1.101	1.667	1.531	1.65	1.136	1.211	1.146

2AG	LT 3015			LT 3017			LT 3114			B3		
5	1.064	1.048	1.022	1.057	1.046	1.095	1.652	1.7	1.631	1.135	1.177	1.201
2.5	1.012	0.976	0.972	1.005	1.057	1.061	1.625	1.604	1.582	1.079	1.175	1.175
1.25	0.938	0.982	0.932	0.998	1.024	1.027	1.659	1.615	1.66	1.108	1.168	1.124
0.625	0.886	0.917	0.917	0.968	1.108	1.029	1.589	1.66	1.608	1.197	1.131	1.139
0.3125	1.021	0.982	0.981	1.047	1.077	1.066	1.626	1.581	1.605	1.142	1.129	1.158
0.15625	1.077	0.959	0.888	1.058	1.105	1.037	1.473	1.586	1.508	1.129	1.138	1.165
0.078125	0.975	1.032	1.063	1.124	1.094	1.117	1.65	1.641	1.664	1.142	1.116	1.21
0	1.119	1.007	1.07	1.112	1.036	1.075	1.747	1.727	1.748	1.251	1.209	1.247

GP	LT 3015			LT 3017			LT 3114			B3		
5	0.989	1.084		1.127	1.139	1.085	1.7	1.674	1.747	1.244	1.245	1.336
2.5	1.044	1.121		1.187	1.131	1.14	1.724	1.747	1.765	1.226	1.237	1.269
1.25	1.087	1.125		1.149	1.154	1.12	1.688	1.83	1.72	1.179	1.247	1.215
0.625	1.097	1.161		1.184	1.087	1.191	1.759	1.784	1.802	1.217	1.248	1.258
0.3125	1.134	1.109		1.127	1.036	1.18	1.83	1.846	1.734	1.204	1.182	1.218
0.15625	1.113	1.113		1.159	1.08	1.12	1.652	1.603	1.694	1.219	1.268	1.275
0.078125	1.117	1.116		1.225	1.175	1.127	1.75	1.693	1.798	1.173	1.309	1.253
0	1.195	1.096		1.216	1.229	1.132	1.789	1.749	1.766	1.279	1.245	1.304